

10th International Conference of Information and Communication Technology (ICICT-2020)

Extraction of Specific Arguments from Chinese Financial News with out-of-domain Samples

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Abstract

The object and evidence are two important parts of financial fraud detection (FFD). In traditional FFD tasks, the detected object generally refers to the enterprise involved in typical financial fraud events, and the evidence comes from corporate statements such as auditing report, etc. Thus, previous FFD methods based on financial statements are limited by the scope of detected objects and the availability of evidence. In this study, we design a financial event library to enlarge the detection scope and expand the source of evidence to financial news, and thus financial entities involved in high-risk events are extracted. The financial event library contains common negative events, each of which corresponds to a negative behaviour that may increase financial risk. Moreover, we propose a novel method to convert the event detection task into the Q&A mode task, which also contributes to the enlargement of the original hand-built dataset. Compared with existing methods on the enlargement of financial dataset, our approach does not require additional annotation. We apply our method into Chinese financial news corpus, and achieves good performance in the extraction task.

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Peer-review under responsibility of the scientific committee of the 10th International Conference of Information and Communication Technology.

Keywords: Financial risk control; Event extraction; Entity extraction; Out-of-domain data; Chinese text analysis

1. Introduction

In recent years, some high-risk financial events have caused huge economic losses, and seriously affected related industries. In these events, financial entities such as financial institutions and enterprises have always played a major role, and their negative behaviors may increase financial risk. Therefore, detecting negative events with potential

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risks and extracting relevant entities can serve as an important basis for further financial analysis to avoid serious losses.

A typical type of negative event is financial fraud. Research on financial fraud^{1,2,3} involves different directions, such as credit card fraud, money laundering, and financial statement fraud, etc. For addressing these problems, financial fraud detection⁴ (FFD) has been proposed and widely used as a classical method of financial risk control. The detected objects and evidence are two important parts of FFD. However, the shortcomings of these two parts limit the performance of traditional FFD: 1) the range of types of events detected is quite limited, because the financial fraud event is the only detected object. Previous methods cannot realize the detection of complex financial fraud events with multiple types of fraud. 2) The sources of evidence are limited and private. Evidence comes from specific types of financial fraud data, including company financial documents^{5,6}, transaction records⁷, and audit reports⁸, etc. All evidence comes from internal and private financial documents, which is difficult to collect⁹.

Compared with FDD tasks, the current goal is no longer to determine whether there is financial fraud, but to detect certain types of negative events and extract the financial entities involved. The contributions of this study mainly include the following three aspects: 1) We have designed an event library to expand the scope of detecting event types, which includes not only classic negative events (e.g. financial fraud), but also some common events with potential risks. 2) Evidence is not limited to the company's financial documents. We use the financial news from the well-known Chinese media as the source of evidence, which contains authoritative and sufficient information from different financial industries. 3) We propose a novel approach based on the traditional event extraction method, which applies a Q&A mode to the extraction task. It brings two obvious advantages. One is that the complexity of the model is reduced. The other is that the original hand-built dataset can be enlarged without additional annotation. The Q&A mode offers the possibility of introducing external dataset directly.

The rest of the paper is organized as follows. Section 2 reviews previous research on event extraction methods and analyzes the advantages and disadvantages. Section 3 presents the framework of the approach, which includes the financial event library, the enlargement method of dataset, and the extraction model. Section 4 introduces the external dataset and provides experimental results. The final conclusions and more discussions are presented in Section 5.

2. Background and related work

Mining specific information from financial documents and other evidences is a key step to identify financial fraud. In our study, the extraction process is divided into two steps: event detection and financial entity extraction. Compared to FFD methods, more types of negative events are considered, and the detected objects are financial entities involved in negative events rather than financial fraud event. Therefore, we adopt a different method, event extraction (EE), to meet the current requirements.

The purpose of event extraction is to identify events of interest in natural language texts. It was first proposed by Automatic Content Extraction (ACE) program¹⁰. An EE task is divided into two parts: the event trigger recognition and event argument recognition, and two typical extraction frameworks are generated accordingly. One is the framework of pipeline, which first recognizes the event trigger, and then extracts the event arguments from the context centered on the trigger words. David Ahn¹¹ proposed a remarkable approach to divide EE into 4 sub-tasks, including event trigger, arguments and argument attributes recognition, and event disambiguating. The other typical framework is the joint extraction, which can recognize event triggers and event arguments at the same time. Chen et al¹² used a dynamic multi-pooling convolutional neural networks (DMCNN) to automatically obtain both lexical-level and sentence-level features. Nguyen et al¹³ showed that the bidirectional LSTM had a better performance in capturing context information of text.

Patterns learning is a combination of extraction rules and domain knowledge, including syntactic rules, semantic rules and dictionaries¹⁴ (e.g. event trigger dictionary, event type dictionary, etc.). Huffman¹⁵ proposed a system called LIEP, which can automatically learn extraction patterns from user-supplied samples. Riloff¹⁶ presented an improved AutoSlog system, which eliminated the dependency on annotated text. In patterns learning, the distant supervision¹⁷ is a common method to enlarge insufficient domain dataset. By distant supervision, EE tasks in any field can easily obtain rich data and enlarge original dataset from open source databases (e.g. Wikipedia, FrameNet¹⁸ and Freebase¹⁹). But the enlarged part of dataset may contain too much noise data, which reduces the learning effect

of the model. Moreover, because external databases are usually large, additional data annotation also means higher cost.

3. Approach

3.1. Financial event library

We design a financial event library which contains two parts to be achieved: 1) A list of common negative financial event types, which is summarized from authoritative financial news. 2) The new event template, which redefines the components of the event and provides a suitable structure for our extraction task.

For obtaining the types of influential and common events, we design a hybrid method that consists of expert knowledge and algorithm of word frequency statistics. The text for word frequency statistics comes from Chinese financial news corpus. We set up filter words to ignore irrelevant or neutral articles, such as policy analysis and expert reviews, and ended up with about 5000 filtered articles. These articles are divided into different categories according to the news sections, including the stock trading section, the bank section and the enterprise section, etc. We then assigned TF-IDF algorithm²⁰ to calculate the word frequency on each category. The calculation methods of *term frequency* (TF) and *inverse document frequency* (IDF) are shown in Eq. (1) and (2), respectively.

$$TF(w_i) = \frac{f(w_i)}{\sum f(w_i)} \quad (1)$$

$$IDF(w_i) = \log \left(\frac{|D|}{DF(w_i)} \right) \quad (2)$$

Where w_i represents a single word and $f(w_i)$ represents the number of times w_i appears in the current document. $|D|$ represents the number of documents, and $DF(w_i)$ is the number of documents involved word w_i . Obviously, if a word appears only in one document, then its corresponding IDF value will reach the maximum value, which indicates that the word has a great influence on the current category. And the final weight $F(w_i)$ of a word in a document is shown in Eq. (3).

$$F(w_i) = TF(w_i) \cdot IDF(w_i) \quad (3)$$

For each document, we rank all the words in the order of $F(w_i)$ value from high to low and reserve the top five words. Then, the results of all documents under each category are integrated, and ten words with the highest $F(w_i)$ value are selected as candidate words for building event types.

As mentioned, the event type defined in our study is in the form of phrase to express more abundant semantic information. Thus, a candidate word that can build the event type need to meet two characteristics: 1) It can be combined with other candidate words to form phrases that have fluent natural language meaning. 2) It is closely related to financial field. To select the appropriate candidate words, we design a candidate word selection algorithm based on expert knowledge. Thus, we provide at least one event type from expert knowledge summary for each category, as an important basis of candidate word selection algorithm. The details of the algorithm are shown in Fig. 1. Finally, we provide 21 types of financial negative events.

Premise: v_i and n_i represent verb and noun in candidate words respectively, v_j and n_j represent verb and noun in event types summarized by expert knowledge respectively. The phrases mentioned in the algorithm contain only one verb and one noun.

Algorithm:

1. If there is any category remained to be processed:
2. Combine v_i and n_i semantically. Note the combination result as p_k if it can describe a certain type of negative event, and note the rest of candidate words as R_i ;
3. When v_j and n_j do not exist, remove R_i and skip to 6.
4. Note the synonyms for v_j or n_j that exist in R_i as s_i , and remove the candidate words in R_i other than s_i ;
5. Expand s_i into the corresponding negative event types according to the events expressed by v_j and n_j , and make the expansion result consistent with the form of p_k ;
6. Move all p_k to the set of final results, and skip to 1.

Fig. 1. The candidate word selection algorithm.

Event template is a common module in an EE task, because it can provide a unified way of expressing structured results. In addition, the event template plays a key role in the establishment of different functional modules of the extraction model. In traditional EE task, event template contains event type, event arguments and trigger words, which divides the extraction model into recognition part and relations inference part. However, in our study, specific financial entities are the only objects to be extracted, as the information of trigger words and event arguments have been expressed in the event types. Thus, we define a new event template, in which the event argument only contains the financial entity. In addition, the financial entity always plays as the action initiator, and thus the attributes of the entities can be obtained without additional inference.

3.2. Enlargement of dataset

In distant supervision methods, the original data will be used as the matching key, and the data in the large knowledge base containing the same key will be added to the original dataset. Studies^{21,22} on distant supervision have shown that the method is effective in data enlargement, however, there are two drawbacks: 1) Additional annotation on large knowledge bases is costly. 2) More noise data will be produced.

For addressing this problem, we propose a novel method that can enlarge the original dataset through out-of-domain dataset without additional annotation. The method is divided into two steps: text conversion and dataset enlargement. The structure of data is similar to that of data from Q&A corpus, which consists of evidence, question and answer. Besides, our task and Q&A task have a similar purpose that to analyze the question and get the answer from the evidence provided. Thus, we can convert the financial entity extraction task into a Q&A task by modifying the event type in event template. The raw text and arguments in the event template can be treated as evidence and answer respectively.

3.3. Model

To improve the understanding ability of the model, we design a model consisting of 3 parts: pre-processing, data combination, and extractor of financial entity. The details are described in Fig. 2.

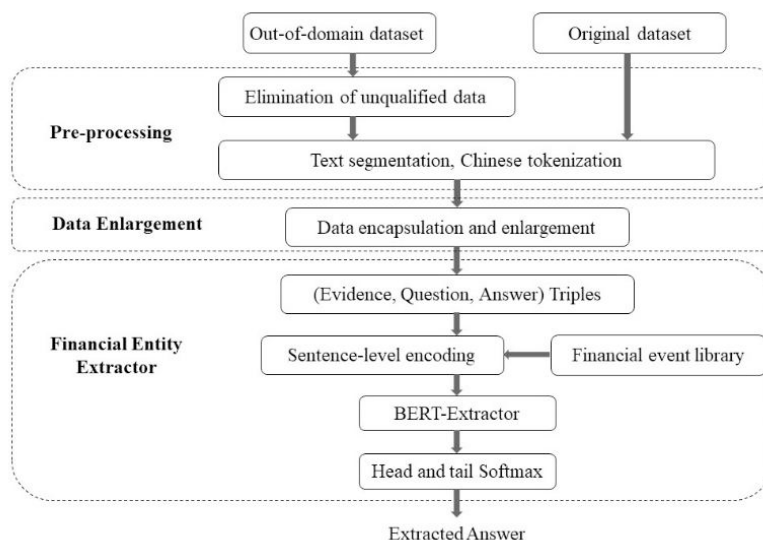


Fig. 2. The model of our extraction task.

Pre-processing

Pre-processing of the input text is the premise of NLP tasks. It contains text segmentation, Chinese tokenization and a data filter. Text segmentation is to segment the long text into fixed length segments to avoid exceeding the processing range of the model. The tokenization of Chinese text contains the word segmentation and the replacement of special punctuation with spaces or special marks. In addition, during the training phase, we will remove unqualified data in the external data set by setting filter parameters such as text length and word count.

Data enlargement

The word enlargement contains a mixture of original and out-of-domain dataset, but also to the combination of extraction task and Q&A task. The text is modified by a fixed sentence pattern, because the repetition of words in sentences of the same type is high. After modification, data from the external dataset and the original dataset is compressed into a triple, (Evidence, Question, and Answer). The triples are arranged out of order so that the data is evenly distributed, and then used as input to the extractor.

Entity extractor

In our study, the Transformer-based model is used as the sentence-level feature extractor. The implementation is actually based on two characteristics of input text: 1) Both evidence and question are represented by natural language, which can maintain sentence-level semantic coherence. 2) The average text length of the evidence or question is not too long, so the text to be processed is always within the detection scope of the model. After the sentence-level representation learning, the answer will be located by two *softmax* functions, one for head word of the entity and the other for the tail word.

4. Experiment

The dataset is divided into two parts: 1) 17815 annotated samples from 3 well-known Chinese news media from different industries. 2) The open source Chinese Q&A corpus, WebQA. The corpus is generated from Chinese encyclopedia of commonsense knowledge and open Q&A texts.

In our experiment, 70% of the data is used as training data, and the remaining 30% serves as test data. And we use BERT as the feature extractor. The epoch is set to 10 and the batch size is 32. The results are shown in Table 1. We use the value of F1 as the basis of measurement. The model of our approach has reached F1 of 87.75% in the test set. By observing and analyzing the failed samples, we found that some failed extraction results are not strictly

inaccurate. When an abbreviation for the same entity is provided, the model can extract the full name of the entity. And the model can extract only one of the answers if there are multiple answers. The results also show that out-of-domain data has improved the generalization ability of the model.

Table 1. Results on the original dataset and mixed dataset..

	Training set	Validation set	F1(%)
Original dataset	14,674	1,602	86.69
WebQA+original dataset	45,896	4,902	87.75

We also find that the accuracy rate is always in a small fluctuation state. The fluctuation may be caused by a mixture of different data. Although we attempted to ensure a balance between the amount of WebQA and original data before experiment, the indexes of training data would be randomly disordered again in each epoch, resulting in an imbalance.

5. Conclusions

The detection of the financial entities, such as enterprise and financial institution, provides a reference for company assessments. In this study, we have established a financial event library for extracting financial entities of negative events from Chinese financial news. We then have developed a framework to encapsulate information of event trigger and event argument into the event type. Compared with traditional FFD task, the scope of detected events is enlarged to negative financial events, and the sources of evidence are also more diverse and abundant.

However, our method has two issues still remained to be addressed: 1) Our method still relies on manual intervention. Although we have summarized 21 typical types from well-known Chinese financial media, the establishment still relies on expert knowledge, and the new common types need to be added manually. 2) The events in financial news may be complicated. The type of event may be implicit, which provides insufficient evidence to recognize the financial entities.

Acknowledgement

This work was supported by the grant of Sichuan Science and Technology Program (No. 2018GZDZX0042), China.

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